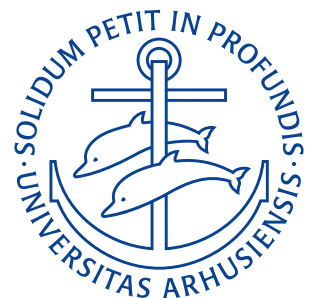


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Exam Preparation

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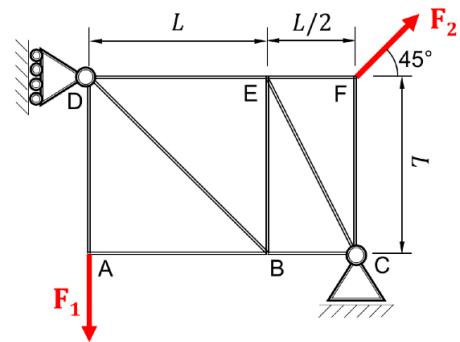
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Lecture 28: 2025 ReExam

12. Januar 2026

Problem 1

Consider the simple truss structure, where the joints are numbered progressively from A to F and the length parameter L is equal to 1,0 m. The truss is supported by a smooth pin at C and a roller at D . Two forces of magnitude $F_1 = 75 \text{ kN}$ and $F_2 = 212,1 \text{ kN}$ are applied at joints A and F , respectively. The direction of these two forces is as indicated on the figure.



Q1.1 Identify all the zero-force members in the truss structure

Q1.2 Determine whether the structure is statically determinate.

Q1.3 Find the reaction forces at C and at D .

Q1.4 Use the method of joints to find the force in member CE .

The truss members are slender bars made of titanium (Young's modulus 120 GPa), with cross-sectional area $A = 625 \text{ mm}^2$.

Q1.5 Calculate the length variation of member CE due to the forces applied to the truss.

Derivation

Q1.1 We immediately realize that AB must necessarily be a zero-force member as it is the only member supplying a force in the x direction at A . This is confirmed to be the only zero force member by a truss analysis in MDSolids as shown on Figure 0.1.

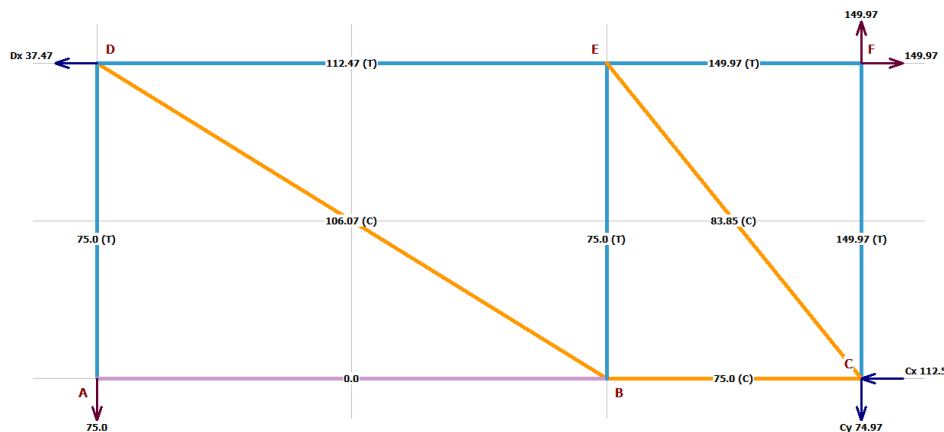


Figure 0.1: Truss analysis in MDSolids

Q1.2 We know that for a statically determinate truss the following equation holds:

$$n_{\text{member}} + n_{\text{reactions}} = 2n_{\text{joints}}.$$

Plugging in numbers (and remembering that C supplies two reactions) we get:

$$9 + 3 = 2 \cdot 6 = 12.$$

which means the structure is statically determinate.

Q1.3 We need to find the reactions D_x , C_x and C_y . Using a force equilibrium in the y direction we find C_y as:

$$\begin{aligned}\sum F_y &= 0 \\ -75 \text{ kN} + C_y + 212,1 \text{ kN} \cdot \sin 45^\circ &= 0 \\ C_y &= 75 \text{ kN} - 212,1 \text{ kN} \cdot \sin 45^\circ = -74,977 \text{ kN}.\end{aligned}$$

A force equilibrium in the x direction gives

$$\begin{aligned}\sum F_x &= 0 \\ D_x + C_x + 212,1 \text{ kN} \cdot \cos 45^\circ &= 0 \\ D_x + C_x &= -149,98 \text{ kN}.\end{aligned}$$

A moment equilibrium about C gives:

$$\begin{aligned}\sum M_C &= 0 \\ \frac{3}{2}L \cdot F_1 - LD_x - LF_2 \cos 45^\circ &= 0 \\ LD_x &= \frac{3}{2}LF_1 - LF_2 \cos 45^\circ \\ D_x &= \frac{3}{2}F_1 - F_2 \cos 45^\circ \\ D_x &= \frac{3}{2} \cdot 75 \text{ kN} - 212,1 \text{ kN} \cdot \cos 45^\circ \\ D_x &= -37,48 \text{ kN}.\end{aligned}$$

Which means that:

$$C_x = -149,98 \text{ kN} + 37,48 \text{ kN} = -112,46 \text{ kN}.$$

This also aligns with the MDSolids simulation on [Figure 0.1](#).

Q1.4 We start by considering joint F . Here, a force equilibrium in the y direction gives

$$\begin{aligned}\sum F_y &= 0 \\ -FC + F_2 \sin 45^\circ &= 0 \\ FC &= F_2 \sin 45^\circ \\ FC &= 212,1 \text{ kN} \cdot \sin 45^\circ = 149,98 \text{ kN}.\end{aligned}$$

Now, a force equilibrium in the y direction at joint C gives us

$$\begin{aligned}\sum F_y &= 0 \\ C_y + FC + CE \cdot \sin 63,435^\circ &= 0 \\ CE &= -\frac{C_y + FC}{\sin 63,435^\circ} \\ CE &= -\frac{(-74,977 \text{ kN}) + 149,98 \text{ kN}}{\sin 63,435^\circ} = -83,355 \text{ kN}.\end{aligned}$$

This also aligns with the MDSolids simulation on [Figure 0.1](#).

Q1.5 We know that the elongation δ caused by an axial force N in a bar of length L , cross-sectional area A and Young's modulus E is

$$\delta = \frac{NL}{AE}.$$

Plugging in numbers we get:

$$\delta = \frac{-83,355 \text{ kN} \cdot 1,12 \text{ m}}{625 \text{ mm}^2 \cdot 120 \text{ GPa}} = -1,245 \text{ mm}.$$

Which means the bar will be compressed by $\delta = 1,245 \text{ mm}$.

Q1.1 $F_{AB} = 0$

Q1.2 Statically Determinate

Q1.3 $C_x = -112,46 \text{ kN}$

$$C_y = -74,98 \text{ kN}$$

$$D_x = -37,48 \text{ kN}$$

Q1.4 $CE = -83,355 \text{ kN}$

Q1.5 $\delta = -1,245 \text{ mm.}$

RESULT

Problem 2

Consider a wooden beam with constant EI loaded as in the figure.

Q2.1 List all the forces, reactions, and kinematic constraints/boundary conditions. Determine whether this structure is statically determinate or indeterminate and explain why.

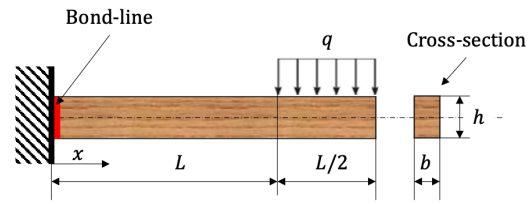
Q2.2 Plot shear force and bending moment diagrams.

Q2.3 Evaluate and plot the elastic curve and calculate the maximum deflection of the beam tip.

Assume that the beam is made of wood with Young's modulus $E = 15 \text{ GPa}$ with a rectangular cross-section of width $b = 50 \text{ mm}$ and thickness $h = 100 \text{ mm}$. $L = 1,5 \text{ m}$. The beam is bonded along to the wall as indicated in the figure.

Q2.4 Evaluate the stress state at three positions along the bond-line and corresponding to: i) the top fibre, ii) the bottom fibre, and iii) the neutral axis.

Q2.5 Draw the Mohr circle and determine the principal stress in the bond-line corresponding to the top fibre.



Derivation

Q2.1 The only applied load is q . The reactions at the bond are R_x , R_y , M_0 . At the fixed end transverse deflection is prevented $v(0) = 0$, so is rotation $v'(0) = 0$ and axial motion $u(0) = 0$. At the free end there is no applied moment so the internal bending moment must be zero $M(3L/2) = 0$, and there is no applied shear so the internal shear is zero $V(3L/2) = 0$, and so is the axial force $N(3L/2) = 0$. This means that we have three unknowns R_x , R_y , and M_0 and three equilibrium equations $\sum F_x = 0$, $\sum F_y = 0$ and $\sum M = 0$, which means the structure is determinate.

Q2.2 The distributed load corresponds to a point load of

$$Q = -\frac{qL}{2}$$

acting directly downwards at $x = 5/4L$. This means that the bond to the wall must supply a reaction force of $R_y = Q = -qL/2$. This means our shear diagram is given by:

$$V(x) = \begin{cases} -\frac{qL}{2} & 0 \leq x \leq L \\ -q(\frac{3}{2}L - x) & L \leq x \leq \frac{3}{2}L. \end{cases}$$

For $0 \leq x \leq L$ the entire loaded patch $L/2$ acts as a resultant $qL/2$ at $x = 5/4L$.

$$M(x) = -\frac{qL}{2} \left(\frac{5}{4}L - x \right), 0 \leq x \leq L.$$

And for $L \leq x \leq 3/2L$ only the remaining tail $3L/2 - x$ is loaded:

$$M(x) = -\frac{q}{2} \left(\frac{3}{2}L - x \right)^2, L \leq x \leq \frac{3}{2}L.$$

A shear and moment diagram is shown on [Figure 0.2](#)

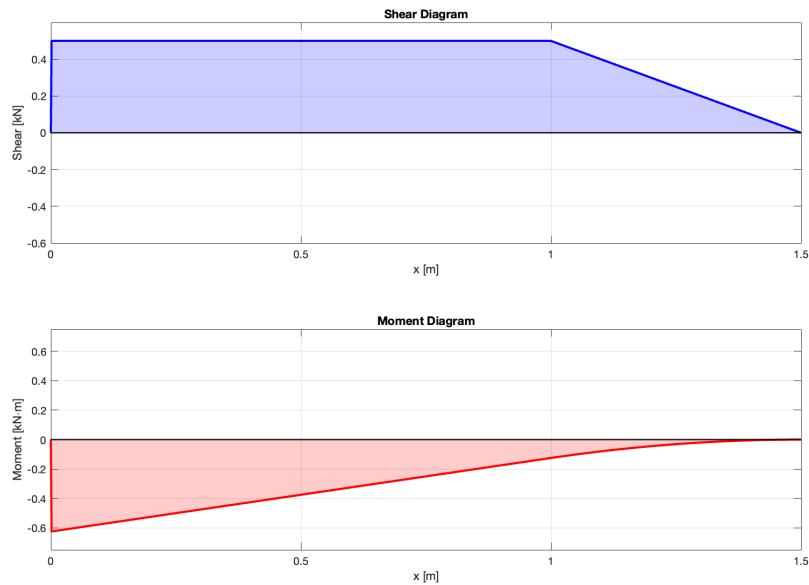


Figure 0.2: Shear and moment diagram.

Q2.3 For an Euler-Bernoulli beam, we have that

$$EIv''(x) = M(x).$$

Which will be integrated piecewise. For $0 < x < L$ we get

$$\begin{aligned} v_1''(x) &= -\frac{qL}{2} \left(\frac{5}{4}L - x \right) \cdot \frac{1}{EI} \\ v_1'(x) &= -\frac{Lq(5Lx - 2x^2)}{8EI} + C_1 \\ v_1(x) &= -\frac{Lqx^2(15L - 4x)}{48EI} + C_1x + C_2. \end{aligned}$$

Here we use that $v(0) = 0 \Rightarrow C_2 = 0$ and as rotation if prevented $v'(0) = 0 \Rightarrow C_1 = 0$ so

$$v_1(x) = -\frac{Lqx^2(15L - 4x)}{48EI}, \quad 0 < x < L.$$

And for $L < x < 3/2L$ we get

$$\begin{aligned} v_2''(x) &= -\frac{q}{2} \left(\frac{3}{2}L - x \right)^2 \cdot \frac{1}{EI} \\ v_2'(x) &= \frac{q(3L - 2x)^3}{48EI} + C_3 \\ v_2(x) &= -\frac{q(3L - 2x)^4}{384EI} + C_3x + C_4. \end{aligned}$$

For continuity at $x = L$ we require $v_1(L) = v_2(L)$ and $v_1'(L) = v_2'(L)$, so,

$$\begin{aligned} v_1(L) &= v_2(L) \\ v_1'(L) &= v_2'(L) \\ -\frac{LqL^2(15L - 4L)}{48EI} &= -\frac{q(3L - 2L)^4}{384EI} + C_3L + C_4 \\ -\frac{Lq(5L^2 - 2L^2)}{8EI} &= \frac{q(3L - 2L)^3}{48EI} + C_3 \end{aligned}$$

$$C_3 \rightarrow -\frac{19L^3 q}{48EI}$$

$$C_4 \rightarrow \frac{65L^4 q}{384EI}.$$

So

$$v(x) = \begin{cases} -\frac{Lqx^2(15L-4x)}{48EI} & 0 \leq x \leq L \\ -\frac{q(3L-2x)^4}{384EI} - \frac{19qL^3}{48EI}x + \frac{65L^4 q}{384EI} & L \leq x \leq \frac{3}{2}L. \end{cases}$$

This is plotted on Figure 0.3.

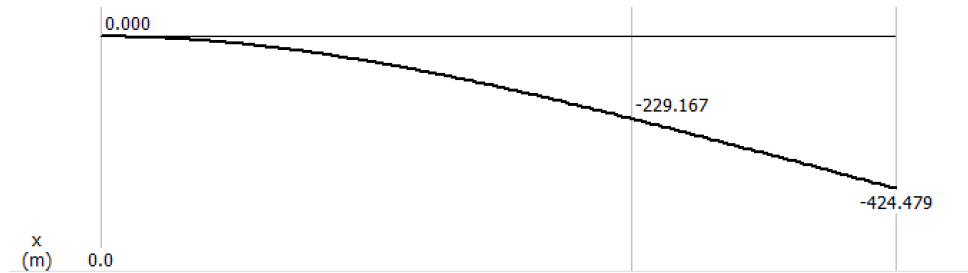


Figure 0.3: Deflection of the beam.

The maximum deflection happens at the beam tip, where the deflection is:

$$\begin{aligned} v\left(\frac{3}{2}L\right) &= -\frac{q(3L-2 \cdot \frac{3}{2}L)^4}{384EI} - \frac{19qL^3}{48EI} \cdot \frac{3}{2}L + \frac{65L^4 q}{384EI} \\ &= -\frac{57qL^4}{96EI} + \frac{65L^4 q}{384EI} \\ &= -\frac{163L^4 q}{384EI}. \end{aligned}$$

Q2.4 The resultants at the fixed end are, as derived above:

$$R_y = \frac{qL}{2}, \quad M_0 = -\frac{qL}{2} \cdot \frac{5}{4}L = -\frac{5qL^2}{8}.$$

We, furthermore, know that

$$\sigma_x = -\frac{M(x)y}{I}$$

and

$$\tau = \frac{V(x)Q(y)}{It}.$$

The second moment of area I of the rectangular cross section is

$$I = \frac{bh^3}{12} = \frac{50 \text{ mm} \cdot (100 \text{ mm})^3}{12} = 4,1667 \cdot 10^6 \text{ mm}^4.$$

For the top fibre we get:

$$\begin{aligned} \sigma_{\text{top}} &= -\frac{M_0 \cdot \frac{h}{2}}{I} \\ &= \frac{5L^2 h}{16 \cdot I} q \\ &= \frac{5 \cdot (1,5 \text{ m})^2 \cdot 100 \text{ mm}}{16 \cdot 4,1667 \cdot 10^6 \text{ mm}^4} q \\ &= 16,874 \cdot 10^3 \text{ m}^{-1} q. \end{aligned}$$

and

$$\tau_{\text{top}} = \frac{V(x)Q(y)}{It}$$

$$\begin{aligned}
 &= -\frac{L \cdot Q}{2I} q \\
 &= -\frac{1,5 \text{ m} \cdot 0 \text{ mm}^3}{2 \cdot 4,1667 \cdot 10^6 \text{ mm}^4} \\
 &= 0.
 \end{aligned}$$

For the bottom fibre ($x = 0$, $y = -h/2$) we get

$$\begin{aligned}
 \sigma_{\text{bot}} &= \frac{M_0 h}{2I} \\
 &= -\frac{5L^2 h}{16I} q \\
 &= -16,874 \cdot 10^3 \text{ m}^{-1} q.
 \end{aligned}$$

and

$$\begin{aligned}
 \tau_{\text{bot}} &= \frac{-\frac{5}{8} q L^2 \cdot 0 \text{ mm}^3}{I b} \\
 &= 0.
 \end{aligned}$$

For the fibre at the neutral axis we get:

$$\begin{aligned}
 \sigma_{NA} &= -\frac{M(x)y}{I} \\
 &= \frac{M(x) \cdot 0 \text{ mm}}{I} \\
 &= 0.
 \end{aligned}$$

and

$$\begin{aligned}
 \tau_{NA} &= \frac{V(x)Q(y)}{I t} \\
 &= -\frac{qL}{2} \cdot \frac{bh}{2} \cdot \frac{h}{4} \cdot \frac{1}{Ib} \\
 &= \frac{Lh^2}{16I} q \\
 &= \frac{1,5 \text{ m} \cdot (100 \text{ mm})^2}{16 \cdot 4,1667 \cdot 10^6 \text{ mm}^4} \\
 &= 225 \text{ m}^{-1} q.
 \end{aligned}$$

Q2.5 At the top fibre we have:

$$\sigma_x = 16,874 \cdot 10^3 \text{ m}^{-1} q \quad \sigma_y = 0 \quad \tau_{xy} = 0.$$

The Mohr's circle is thus given by

$$\begin{aligned}
 \sigma_{\text{avg}} &= \frac{\sigma_x + \sigma_y}{2} = \frac{16,874 \cdot 10^3 \text{ m}^{-1} q}{2} = 8,437 \cdot 10^3 \text{ m}^{-1} q \\
 R &= \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \sqrt{\left(\frac{16,874 \cdot 10^3 \text{ m}^{-1} q}{2}\right)^2} = 8,437 \cdot 10^3 \text{ m}^{-1} q.
 \end{aligned}$$

The principal stresses are given by:

$$\sigma_{1,2} = \sigma_{\text{avg}} \pm R \implies \sigma_1 = \sigma_x, \sigma_2 = 0.$$

This circle is shown on [Figure 0.4](#).

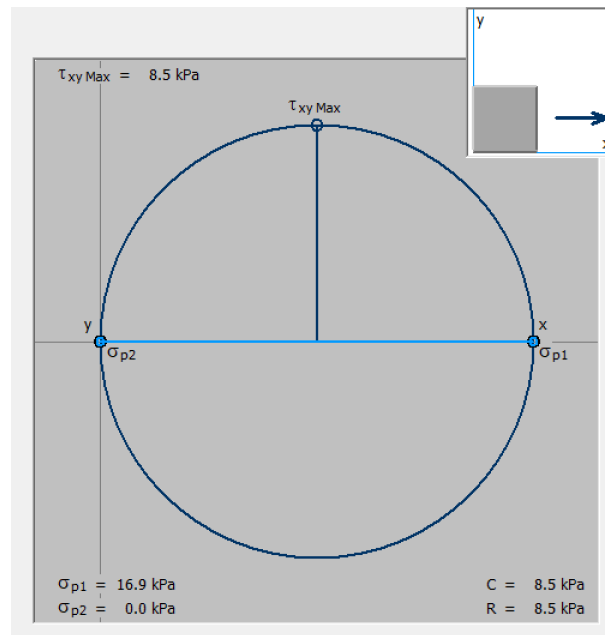


Figure 0.4: Mohr's circle for the stress state at the top-fibre.

Q2.1 See the section

Q2.2 See Figure 0.2

Q2.3 See Figure 0.3

$$v\left(\frac{3}{2}L\right) = -\frac{163}{384EI}L^4q$$

Q2.4 $\sigma_{\text{top}} = 16,874 \cdot 10^3 \text{ m}^{-1}q$

$$\tau_{\text{top}} = 0$$

$$\sigma_{\text{bot}} = -16,874 \cdot 10^3 \text{ m}^{-1}q$$

$$\tau_{\text{bot}} = 0$$

$$\sigma_{NA} = 0$$

$$\tau_{NA} = 225 \text{ m}^{-1}q$$

Q2.5 $\sigma_1 = \sigma_x$

$$\sigma_2 = 0$$

See Figure 0.4.

RESULT

Lecture 28: 2024 Exam

12. Januar 2026

Problem 1

Consider the situation shown in the figure, where a triangular steel plate (in yellow) is connected to a wall via a ball and socket joint at point A . The weight of the plate is 500 N, and the gravity force acts downward along the negative y axis. To maintain equilibrium, the plate is also connected to two cables at points B and C . The thickness of the plate can be assumed uniform and negligible in relation to the other geometrical dimensions. Therefore, points A , B and C can be considered to all lie on the z axis,

Q1.1 Draw a free body diagram of the plate indicating all the forces acting on it.

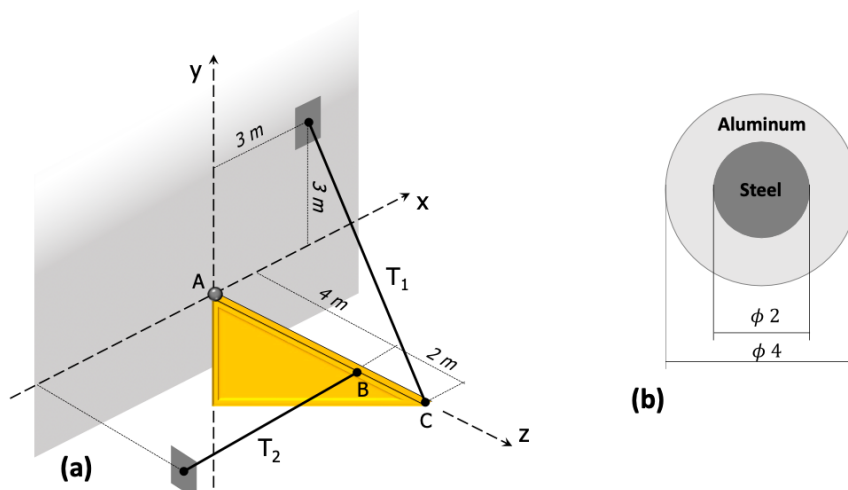
Q1.2 Compute the unit vector along which the tension T_1 acts (directed from point C along the cable)

Q1.3 Use moment equilibrium to find the magnitude of the tension T_1 .

Q1.4 Use the remaining equilibrium equations to find all the other unknown forces acting on the plate.

Assume now that the cable connected at point C can be considered a slender bar with circular cross-section, consisting of a steel core and an aluminum outer shell. The outer diameter of the cable is 4 mm, and the diameter of the steel core is 2 mm.

Q1.5 Calculate the elongation of the cable due to the tension T_1 . The Young's modulus of steel and of aluminum can be assumed to be 210 GPa and 70 GPa, respectively.

**Derivation**

Q1.1 Two free-body diagrams (one in the yz plane and one in the xz plane) have been constructed and they can be seen on [Figure 0.5](#).

Q1.2 T_1 acts along the line from $C = (0; 0; 6\text{m})$ to $D = (3\text{m}; 3\text{m}; 0)$. This means that it acts along the vector:

$$\mathbf{CD} = \mathbf{D} - \mathbf{C} = (3\text{m}; 3\text{m}; -6\text{m}).$$

A unit vector along this can now be found by dividing $\mathbf{CD}$ by $|\mathbf{CD}|$ as:

$$\begin{aligned}\hat{u}_{T_1} &= \hat{u}_{\mathbf{CD}} \\ &= \frac{(3\text{m}; 3\text{m}; -6\text{m})}{\sqrt{(3\text{m})^2 + (3\text{m})^2 + (6\text{m})^2}}\end{aligned}$$

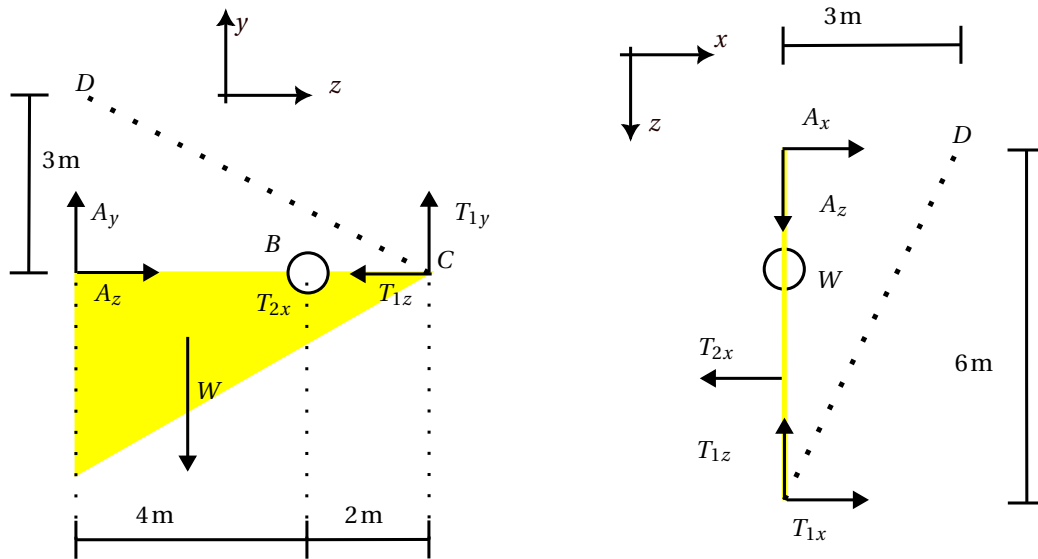


Figure 0.5: Free body diagram.

$$= (0,408; 0,408; -0,816)$$

$$= 0,408\mathbf{i} + 0,408\mathbf{j} - 0,816\mathbf{k}.$$

Q1.3 We use a moment equilibrium about point A in the yz direction to obtain. Note that the center of gravity of the steel plate will lie at $z = 2\text{ m}$ as it is a triangle. Hence:

$$\begin{aligned}\sum (M_A)_{yz} &= 0 \\ T_{1y} \cdot 6\text{ m} - W \cdot 2\text{ m} &= 0 \\ T_{1y} &= \frac{500\text{ N} \cdot 2\text{ m}}{6\text{ m}} \\ &= 166,67\text{ N}.\end{aligned}$$

Using the unit vector we can then find the magnitude of T_1 as

$$\mathbf{T}_1 = \frac{T_{1y}}{(\hat{u}_{T_1})_y} = \frac{166,67\text{ N}}{0,408} = 408,5\text{ N}.$$

Q1.4 Using a global force equilibrium in the y direction we obtain the reaction A_y as:

$$\begin{aligned}\sum F_y &= 0 \\ A_y - W + F_{1y} &= 0 \\ A_y &= -166,67\text{ N} + 500\text{ N} \\ A_y &= 333,33\text{ N}.\end{aligned}$$

A force equilibrium in the z direction yields:

$$\begin{aligned}\sum F_z &= 0 \\ A_z - F_{1z} &= 0 \\ A_z &= F_{1z} \\ A_z &= 408,5\text{ N} \cdot -0,816 \\ A_z &= -333,336\text{ N}.\end{aligned}$$

And in the x direction we get:

$$\begin{aligned}\sum F_x &= 0 \\ A_x - T_2 + T_{1x} &= 0 \\ A_x - T_2 &= -T_{1x} \\ A_x - T_2 &= -408,5 \text{ N} \cdot 0,408 \\ A_x - T_2 &= -166,668 \text{ N}.\end{aligned}$$

And now we use a moment equilibrium about A in the xz direction to get:

$$\begin{aligned}\sum (M_A)_{xz} &= 0 \\ -T_2 \cdot 4 \text{ m} + T_{1x} \cdot 6 \text{ m} &= 0 \\ T_2 &= \frac{T_{1x} \cdot 6 \text{ m}}{4 \text{ m}} \\ T_2 &= \frac{166,67 \text{ N} \cdot 6 \text{ m}}{4 \text{ m}} \\ T_2 &= 250 \text{ N}.\end{aligned}$$

Which means that:

$$A_x = -166,668 \text{ N} + 250 \text{ N} = 83,332 \text{ N}.$$

I.e.

$$\begin{aligned}A_x &= 83,33 \text{ N} \\ A_y &= 333,33 \text{ N} \\ A_z &= -333,33 \text{ N} \\ T_2 &= 250 \text{ N}.\end{aligned}$$

Q1.5 For there to be no sliding between the aluminum and steel layers we require $\epsilon_{st} = \epsilon_{al} = \epsilon$. Hence,

$$\sigma_{st} = E_{st}\epsilon \quad \sigma_{al} = E_{al}\epsilon.$$

The force is then

$$T_1 = \sigma_{st} \cdot A_{st} + \sigma_{al} \cdot A_{al} = \epsilon (E_{st}A_{st} + E_{al}A_{al}).$$

So,

$$\begin{aligned}\epsilon &= \frac{T_1}{E_{st} \cdot \pi \cdot \frac{(2 \text{ mm})^2}{4} + E_{al} \cdot \pi \cdot \frac{(4 \text{ mm})^2 - (2 \text{ mm})^2}{4}} \\ &= \frac{408,5 \text{ N}}{210 \text{ GPa} \cdot \pi \cdot \frac{(2 \text{ mm})^2}{4} + 70 \text{ GPa} \cdot \pi \cdot \frac{(4 \text{ mm})^2 - (2 \text{ mm})^2}{4}} = 3,096 \cdot 10^{-4}.\end{aligned}$$

So the elongation is:

$$\delta = |CD|\epsilon = \sqrt{9 \text{ m}^2 + 9 \text{ m}^2 + 36 \text{ m}^2} \cdot 3,096 \cdot 10^{-4} = 2,28 \text{ mm}.$$

Q1.1	See Figure 0.5
Q1.2	$\hat{u}_{T_1} = 0,408\mathbf{i} + 0,408\mathbf{j} - 0,816\mathbf{k}$
Q1.3	$T_1 = 408,5\text{N}$
Q1.4	$A_x = 83,33\text{N}$ $A_y = 333,33\text{N}$ $A_z = -333,33\text{N}$ $T_2 = 250\text{N}$
Q1.5	$\delta = 2,28\text{mm}$.

RESULT

Problem 2

Consider a steel T-section overhand beam loaded as shown in Fig. 2.

Let $L = 4\text{ m}$, $l = 1\text{ m}$, $w_0 = 4 \frac{\text{kN}}{\text{m}}$, $E = 200\text{ GPa}$, $t = 10\text{ mm}$, $f = 15\text{ mm}$, $b = h = 300\text{ mm}$.

- Q2.1 List all reactions and kinematic constraints, establish whether the structure is statically determinate or indeterminate, and find its degree of indeterminacy.
- Q2.2 Find the reaction forces and derive expressions for the shear force and bending moment along the beam.
- Q2.3 On the cross section located at $x = L/2$, draw the Mohr circle at point C and find the principal stresses and directions. (You may want to start by finding the state of stress at point C).

We replace the support at A by a fixed support – See Fig. 3.

- Q2.4 Find the degree of indeterminacy of the structure and compute the reactions at A and B.

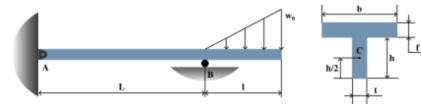


Fig. 2. A steel T-section beam with distributed load in the overhang region.

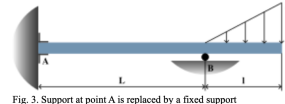


Fig. 3. Support at point A is replaced by a fixed support

Derivation

- Q2.1 As A is a pin it has reactions A_x , A_y and the roller B contributes a reaction of B_y . We have the boundary condition at A that there can be no translation here, hence $v(0) = 0$ and the same is true at B $v(L) = 0$. As we have three unknowns (A_x , A_y and B_y) and three equilibrium equations ($\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$) the structure is statically determinate.
- Q2.2 We start by replacing the distributed load by a corresponding point loading. As the distributed load is triangular the point load will have a magnitude of

$$W_0 = w_0 L = \frac{1}{2} \cdot 4 \frac{\text{kN}}{\text{m}} \cdot 1\text{ m} = 2\text{ kN}.$$

This will act directly downwards at $x = L + 2l/3 = 4\text{ m} + 0,666\text{ m} = 4,666\text{ m}$. A force equilibrium in the y direction then gives:

$$\begin{aligned}\sum F_y &= 0 \\ A_y + B_y - W_0 &= 0 \\ A_y + B_y &= W_0 \\ A_y + B_y &= 2\text{ kN}.\end{aligned}$$

And a moment equilibrium about A yields:

$$\begin{aligned}\sum M_A &= 0 \\ B_y \cdot L - W_0 \cdot 4,666\text{ m} &= 0 \\ B_y &= \frac{W_0 \cdot 4,666\text{ m}}{L} \\ B_y &= \frac{2\text{ kN} \cdot 4,666\text{ m}}{4\text{ m}} \\ B_y &= 2,333\text{ kN}.\end{aligned}$$

Which means

$$A_y = 2 \text{ kN} - 2,333 \text{ kN} = -0,333 \text{ kN}.$$

Hence, our shear diagram is defined by:

$$V(x) = \begin{cases} A_y, & 0 \leq x \leq L \\ A_y + B_y - \int_L^x w(s) \, ds, & L \leq x \leq L + l. \end{cases}$$

We know that $A_y + B_y = W = 2 \text{ kN}$ and $w(s) = \frac{w_0}{l}(s - L)$ So

$$\begin{aligned} \int_L^x w(s) \, ds &= \int_L^x \frac{w_0}{l}(s - L) \, ds \\ &= \frac{w_0}{2l}(x - L)^2 \\ \frac{w_0 l}{2} - \frac{w_0}{2l}(x - L)^2 &= \frac{4 \frac{\text{kN}}{\text{m}} \cdot 1 \text{ m}}{2} - \frac{4 \frac{\text{kN}}{\text{m}}}{2 \cdot 1 \text{ m}}(x - 4 \text{ m})^2 \\ V(x) &= \begin{cases} -0,333 \text{ kN} & 0 \leq x \leq L \\ 2 \text{ kN} - 2 \text{ kN/m}^2 (x - 4 \text{ m})^2 & L \leq x \leq L + l. \end{cases} \end{aligned}$$

We use $M'(x) = V(x)$ and $M(0) = 0$ to get

$$M(x) = \begin{cases} A_y x, & 0 \leq x \leq L \\ A_y L + \int_L^x V(s) \, ds & L \leq x \leq L + l. \end{cases}$$

Where we get:

$$\begin{aligned} \int_L^x V(s) \, ds &= \int_4^x 2 - 2(s - 4)^2 \, ds \\ &= -\frac{2x^3}{3} + 8x^2 - 30x + \frac{104}{3}. \end{aligned}$$

Which means:

$$\begin{cases} 0,333x & 0 \leq x \leq L \\ -0,333x - \frac{2x^3}{3} + 8x^2 - 30x + \frac{104}{3} & . \end{cases}$$

These are plotted on [Figure 0.6](#)

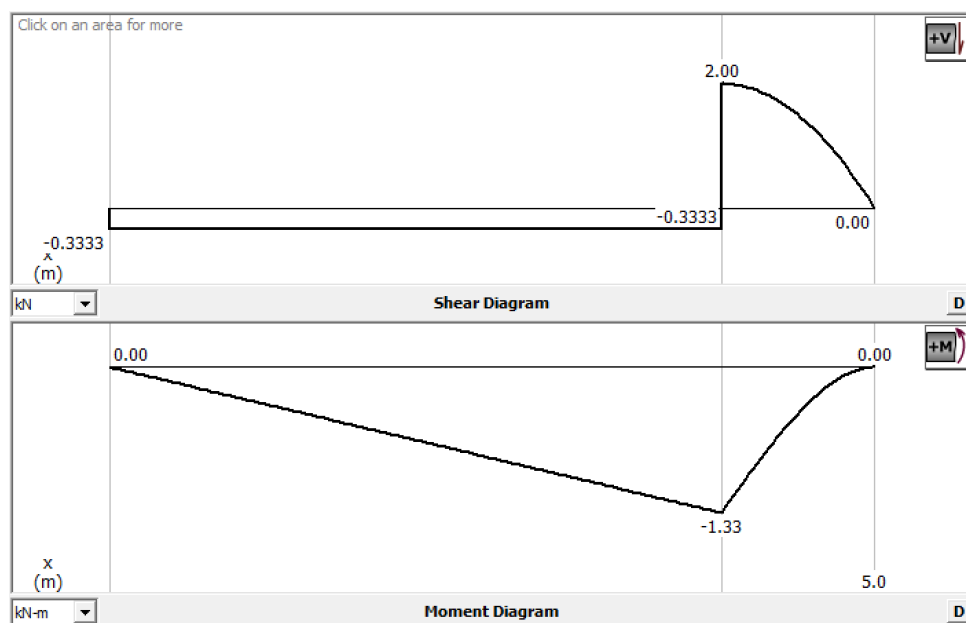


Figure 0.6: Shear and Moment diagram.

Q2.3

Q2.4 Tissemand